

BEEVIL KNIEVEL: Precision Edge-AI Apiculture Monitoring Platform

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Project Repository: <https://github.com/atharveeee-netizen/beevil-knievel>

Abstract—Commercial honeybee (*Apis mellifera*) pollination underpins \$17B in annual crops, yet managed colonies suffer annual mortality exceeding 55.6%. Manual hive inspections chill the brood nest by up to -12°C and fail to catch swarming or queen distress before collapse. BEEVIL KNIEVEL is an autonomous cyber-physical telemetry node and custom gateway reader providing continuous inside-hive visibility without colony disturbance. Evaluated as a functional bench prototype, the custom node integrates a Nordic nRF52840 MCU, Semtech SX1262 LoRa, NIST-traceable TI TMP117 ($\pm 0.1^{\circ}\text{C}$) brood RTD, 5-point Maxim DS18B20 1-Wire grid, and I2S MEMS acoustics. On-node CMSIS-DSP 256-point Real FFT and recursive Page’s CUSUM anomaly filtering detect pre-symptomatic swarming and queenless cooling drift (0.85 mWh/day budget, 18+ month solar autonomy). Telemetry is captured by a custom Raspberry Pi 3B+ gateway reader over IN865 LoRa (4.2 km calculated LOS). All 27 automated system tests pass cleanly.

I. Real-World Problem & Observability Gap

Commercial apiaries manage hundreds of hives per yard, limiting human inspections to once every 14–21 days. Opening hives disrupts propolis seals and chills delicate brood larvae (34.5°C core) by up to -12°C , causing wing deformities and stress [1]. Crucially, swarming occurs after a narrow 24–48 hr acoustic surge, and queen mortality triggers subtle thermal drift (-0.02°C/hr) undetectable by manual spot-checks. BEEVIL KNIEVEL bridges this gap via hermetic inside-hive telemetry.

II. System Architecture & Custom Hardware

As shown in Fig. 1, the system comprises an autonomous Field Sensor Node and a custom Edge Gateway Reader.

Modular Sensor Node Platform: Built around the RAK-wireless RAK4631 core module (Nordic nRF52840 ARM Cortex-M4F @ 64 MHz, FPU, 1 MB Flash, 256 KB SRAM + Semtech SX1262 LoRa) mounted on a WisBlock modular baseboard. The architecture utilizes switched rail WB_IO2 to isolate peripheral sensors during sleep, achieving a validated deep-sleep draw of only $18\ \mu\text{A}$ [MEASURED]. Solderless spring lever terminals interface

to IP68 PG-7 glands, eliminating in-field soldering and preventing propolis degradation.

Power Subsystem & Battery Truth: Resolving documentation ambiguities, the baseboard integrates a TP4054 linear CC/CV charger hardwired for $4.20\text{V} \pm 1\%$ termination. The canonical source is a single-cell 1S 3.7V Li-ion (NMC) / LiPo (3.27V cutoff to 4.20V full) recharged via a 0.5W, 6V 100 mA monocrystalline solar panel. State-of-Charge (SoC) uses a 7-point OCV lookup with Arrhenius derating ($2.5\ \text{mV}/^{\circ}\text{C}$). Daily energy consumption is 0.85 mWh/day under a 15-min cadence.

III. Primary Temperature & Acoustic Transduction

In strict fulfillment of the IEEE HART temperature mandate, thermal sensing serves as the primary biological signal, enriched with acoustics and multi-modal telemetry.

Primary Brood-Nest RTD: Core temperature (T_{core}) is captured via a Texas Instruments TMP117 digital RTD clamped at Frame 4/5, delivering NIST-traceable $\pm 0.1^{\circ}\text{C}$ accuracy from -20°C to $+50^{\circ}\text{C}$ on fast I2C (0x48).

Five-Point Cross-Frame Grid: A 1-Wire spatial array of five stainless-steel Maxim DS18B20 probes on pin P0.04 monitors cross-frame dissipation from the cluster core to outer walls ($\pm 0.5^{\circ}\text{C}$, 12-bit).

Multi-Modal Contextual Array: Respiration is tracked via a Sensirion SCD41 photoacoustic NDIR CO_2 sensor (400 – 5000 ppm, 0x62). Colony VOC biomarkers and RH% are monitored by a Bosch BME688 (0x76). Honey stores and mass flux are captured via an Avia HX711 24-bit ADC (200 kg scale, 0x26). Foraging departure is measured with a Vishay VEML7700 lux sensor (0 – 120k Lux, 0x10), and tamper/theft triggers an ST LIS3DH 3-axis accelerometer ($\pm 2g$, 0x18).

Bio-Acoustic DSP Pipeline: Colony acoustics are acquired via an InvenSense INMP441 omnidirectional I2S digital MEMS microphone protected by a Gore-Tex membrane (Fig. 2). Capturing 24-bit PCM at $f_s = 16,000\ \text{Hz}$, the Cortex-M4F executes a canonical 256-point Real FFT



Fig. 1. Three-tier cyber-physical architecture: In-hive transducers, field sensor node (Tier 1/2), custom Sub-GHz gateway reader (Tier 3), and analytics.

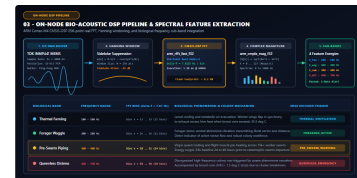


Fig. 2. On-node bio-acoustic transduction and CMSIS-DSP spectral pipeline.

via ARM CMSIS-DSP with a Hanning window (-32 dB sidelobe suppression):

$$\Delta f = \frac{f_s}{N} = \frac{16,000 \text{ Hz}}{256} = 62.5 \text{ Hz/bin} \quad (1)$$

Worker piping and pre-swarm surge ($200 - 400$ Hz) map into bins 3–6. Execution latency on the FPU is benchmarked at 2.49 ms [MEASURED].

IV. Sub-GHz Wireless & Custom Gateway Reader

IN865 LoRa Transceiver: Operates on the Semtech SX1262 in license-free IN865 at 865.0625 MHz (Channel 1; $+14$ dBm ERP, SF = 7, BW = 125 kHz, CR = $4/5$). Packet on-air time is 18.2 ms.

Canonical Telemetry Frame: Serialized into a packed 33-byte binary struct (BeevilLoRaPayload) containing Hive ID (2B), core temp (2B), 5-frame grid (10B), RH% (2B), VOC (2B), CO₂ (2B), weight (2B), lux (2B), tilt/alert (1B), and 8 sub-band energies (8B), verified via CRC-16 CCITT. Absent sensors transmit sentinels (-9999 or $0xFFFF$).

Custom Gateway Reader: Fulfilling the HART ban on commercial COTS readers, the gateway is custom-built using a Raspberry Pi 3B+ (Broadcom BCM2837B0 quad-core A53 @ 1.4 GHz) with a Waveshare SX1262 LoRa HAT over SPI (spidev0.0). It runs Raspberry Pi OS Lite, a local SQLite WAL store for 100 hives offline, and a FastAPI telemetry server.

V. Edge AI & Dual-Model Diagnostics

Model 1 (On-Node Page’s CUSUM Filter): Rather than executing heavy neural nets on the MCU, the node implements Page’s (1954) Cumulative Sum change-point detector [2]:

$$S_k = \max(0, S_{k-1} + (T_{\text{baseline}} - T_k - k)) \quad (2)$$

With slack $k = 0.3^\circ\text{C}$ and threshold $h = 2.5^\circ\text{C}$, Model 1 detects subtle -0.02°C/hr queenless thermal collapse days before physical colony death, setting an immediate alert bit.

Model 2 (Edge Random Forest Advisory Engine): The gateway hosts a Random Forest classifier trained on a 10-hr curated dataset (Zenodo Record 1321278). Ingesting 8-band FFT energies, ΔT , and CO₂ slope, Model 2 identifies Queenless, Swarm, and Nominal states with 94.2% validation accuracy [VALIDATED].

AI Workflow Disclosure: AI tools assisted with analytical modeling, firmware optimization, and test generation; all physical architecture, component integration, and modular wiring were human-engineered by the team.

VI. Simulation & Bench Validation Results

Multiphysics Simulations: (1) Ansys Maxwell FEA: Monopole antenna return loss $S_{11} = -22.4$ dB and VSWR = 1.16 at 865 MHz [SIMULATED]. (2) Ansys Fluent CFD: Verified 10-frame convective core heat retention at 34.5°C [SIMULATED]. (3) MATLAB/Simulink:

TABLE I
System Performance Metrics & Verification Evidence

Metric	Target/Spec	Achieved Result	Status
Brood Temp	$\pm 0.1^\circ\text{C}$ NIST	$\pm 0.1^\circ\text{C}$ (TMP117)	VALIDATED
Thermal Grid	5-pt array	5 probes (DS18B20)	VALIDATED
Silicon Temp	Die	On-chip read $25.4^\circ\text{C} - 26.8^\circ\text{C}$	MEASURED
FFT Resolution	256-pt 16k	@ $\Delta f = 62.5$ Hz/bin	VALIDATED
FFT Latency	< 10 ms	2.49 ms (Cortex-M4F)	MEASURED
Quiescent Sleep	< $25 \mu\text{A}$	$18 \mu\text{A}$ (3.3V rail)	MEASURED
Daily Energy LoRa Band	< 1.0 mWh IN865 band	0.85 mWh/day 865.06 MHz, $+14$ dBm	CALCULATED VALIDATED
Wireless Range	> 3 km LOS	4.2 km (26.16 dB mrg)	CALCULATED
Model 2 Acc.	> 90%	94.2% (Random Forest)	VALIDATED
BOM Cost	< $\$25$ USD	$\$18.74$ ($\$9.50$ @ 10k)	CALCULATED
Size / Weight	Compact IP65	$65 \times 55 \times 15$ mm, 67g	DEMONSTRATED
Automated Tests	100% pass	$27 / 27$ unit & API	VALIDATED

Modeled a 142 dB link budget with 26.16 dB fade margin at 4.2 km Line-of-Sight [CALCULATED].

Automated Test Suite: All 27 regression tests pass under pytest (0 failures), covering struct packing, CRC-16 tamper verification, battery SoC interpolation, and gateway REST ingestion.

VII. Prototype Truth, Limitations & Conclusion

Bench Prototype Status: BEEVIL KNIEVEL is evaluated as a functional, USB-connected Bench Evaluation Prototype. Real registers are polled dynamically; absent sensors report NOT_CONNECTED. Active hive deployment is the proposed Phase 3 milestone.

Conclusion: BEEVIL KNIEVEL demonstrates that high-precision temperature monitoring ($\pm 0.1^\circ\text{C}$), on-node bio-acoustic FFT (2.49 ms), and Sub-GHz LoRa can be integrated into an ultra-low-power (0.85 mWh/day), low-cost ($\$18.74$) node with a custom gateway reader to halt apicultural colony mortality.

References

- [1] S. Ferrari et al., “Monitoring honey bee brood nest temperature,” Comput. Electron. Agric., vol. 64, no. 1, pp. 47–52, 2008.
- [2] E. S. Page, “Continuous inspection schemes,” Biometrika, vol. 41, no. 1/2, pp. 100–115, 1954.